

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier
from sklearn.metrics import accuracy_score, classification_report
```

```
%matplotlib inline
```

```
data = pd.read_csv("fruit_basket_bagging.csv")
```

```
print(data.shape)
data.head()
```

```
(120, 6)
```

	Sample_ID	Weight_g	Sweetness	Firmness	Color_Score	Fruit
0	1	109.5	7.8	4.7	5.5	Banana
1	2	258.9	8.7	6.0	7.8	Mango
2	3	151.7	6.2	6.2	8.5	Orange
3	4	117.6	8.3	6.5	7.2	Banana
4	5	164.6	6.9	6.7	6.5	Apple

```
print(data["Fruit"].value_counts())
```

```
Fruit
Apple      49
Banana     32
Orange     22
Mango      17
Name: count, dtype: int64
```

```
X = data[
    ["Weight_g", "Sweetness", "Firmness", "Color_Score"]
]
```

```
y = data["Fruit"]
```

```
print("Input variables:")
print(X.head())
```

```
print("\nTarget variable:")
print(y.head())
```

```
Input variables:
```

	Weight_g	Sweetness	Firmness	Color_Score
0	109.5	7.8	4.7	5.5
1	258.9	8.7	6.0	7.8
2	151.7	6.2	6.2	8.5
3	117.6	8.3	6.5	7.2

4	164.6	6.9	6.7	6.5
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Target variable:

0 Banana

1 Mango

2 Orange

3 Banana

4 Apple

Name: Fruit, dtype: str

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

```
print("Training samples:", len(X_train))
```

```
print("Testing samples:", len(X_test))
```

Training samples: 96

Testing samples: 24

```
dt = DecisionTreeClassifier(
    random_state=42
)
```

```
dt.fit(X_train, y_train)
```

```
y_pred = dt.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Decision Tree Accuracy:", accuracy)
```

Decision Tree Accuracy: 0.7916666666666666

```
bagging = BaggingClassifier(
    estimator=DecisionTreeClassifier(random_state=42),
    n_estimators=10,
    random_state=42
)
```

```
bagging.fit(X_train, y_train)
```

```
y_pred_bagging = bagging.predict(X_test)
```

```
accuracy_bagging = accuracy_score(
    y_test,
    y_pred_bagging
)
```

```

)

print("Bagging Accuracy:", accuracy_bagging)

Bagging Accuracy: 0.875

results = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred_bagging
})

print(results.head(15))

```

	Actual	Predicted
0	Banana	Banana
1	Mango	Mango
2	Banana	Banana
3	Orange	Orange
4	Apple	Apple
5	Apple	Apple
6	Apple	Apple
7	Apple	Banana
8	Apple	Apple
9	Banana	Apple
10	Orange	Orange
11	Orange	Orange
12	Apple	Apple
13	Apple	Apple
14	Orange	Orange

```

tree_predictions = []

for tree in bagging.estimators_:
    prediction = tree.predict(new_fruit)[0]
    tree_predictions.append(prediction)

print("Individual Tree Predictions:")

for i, prediction in enumerate(tree_predictions):
    print("Tree", i + 1, ":", prediction)

```

Individual Tree Predictions:

```

Tree 1 : 0
Tree 2 : 0
Tree 3 : 0
Tree 4 : 0
Tree 5 : 0
Tree 6 : 0
Tree 7 : 0
Tree 8 : 0

```

```

Tree 9 : 0
Tree 10 : 0

from collections import Counter

vote_count = Counter(tree_predictions)

print("Votes:")
print(vote_count)

final_class = vote_count.most_common(1)[0][0]

print("Final Class:", final_class)

Votes:
Counter({np.int64(0): 10})
Final Class: 0

print(
    classification_report(
        y_test,
        y_pred_bagging
    )
)

```

	precision	recall	f1-score	support
Apple	0.82	0.90	0.86	10
Banana	0.80	0.67	0.73	6
Mango	1.00	1.00	1.00	3
Orange	1.00	1.00	1.00	5
accuracy			0.88	24
macro avg	0.90	0.89	0.90	24
weighted avg	0.87	0.88	0.87	24

INFERENCE:

Bagging was applied using multiple Decision Trees. Different bootstrap samples were used to construct individual trees, and their predictions were combined using majority voting.

The final class is determined by the fruit receiving the highest number of votes from the decision trees.

In the given figure, Apple receives the majority of votes from the individual trees. Therefore, the final class is Apple.